19/03/2026

DAY-4

BLOCK-1 : Aptitude

1.) A can complete a work in 10 days and B in 15 days. How long will they take together?

Solu:

$$A + B = x$$

$$\frac{1}{x} = \frac{1}{10} + \frac{1}{15}$$

$$\frac{1}{x} = \frac{3}{30} + \frac{2}{30} = \frac{5}{30}$$

$$\frac{1}{x} = \frac{1}{6}$$

$$x = 6 \text{ days.}$$

2.) A, B and C can complete a work in 6, 8 and 12 days respectively. How many days will they take working together?

Solu:

A = 6 days, B = 8 days, C = 12 days.

$$A + B + C = \frac{1}{6} + \frac{1}{8} + \frac{1}{12}$$

$$= \frac{4}{24} + \frac{3}{24} + \frac{2}{24} = \frac{9}{24}$$

$$= \frac{3}{8}$$

$$= \frac{8}{3}$$

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3) A alone can do a work in 12 days. After 4 days, B joins and completes remaining work in 4 days. Find time taken by B alone.

Soln:

A's work $= \frac{1}{12}$ per day.

work done in 4 days $= 4 \times \frac{1}{12} = \frac{1}{3}$

Remaining work $= 1 - \frac{1}{3} = \frac{3-1}{3} = \frac{2}{3}$.

A+B complete $\frac{2}{3}$ work in 4 days.

1 Day work $= \frac{2}{3}/4 = \frac{\frac{2}{3}}{\frac{4}{1}} \Rightarrow \frac{2}{3} \times \frac{1}{4} = \frac{1}{6}$.

$A + B = \frac{1}{6}$.

$A = \frac{1}{12}$.

$B = \frac{1}{6} - \frac{1}{12} = \frac{1}{12} \Rightarrow 12$ Days.

4) If 5 workers can complete a work in 8 days. how many workers are required to complete it in 4 days?

Soln:

workers = 5, Days = 8

work = workers × Days

$= 5 \times 8$

$= 40$.

workers $= \frac{\text{work}}{\text{Days}} = \frac{40}{4} = 10$.

= 10 workers

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5.) A pipe fills a tank in 6 hours and another empties it in 8 hours. How long to fill the tank.

Solu:-

Filling pipe $= \frac{1}{6}$

Emptying pipe $= \frac{1}{8}$

$$\frac{1}{6} - \frac{1}{8} = \frac{4}{24} - \frac{3}{24} = \frac{4-3}{24}$$

$$= \frac{1}{24}$$

24 hours. fill the tank.

BLOCK-2: Verbal Ability:-

1.) why are optimized systems important?

Optimized systems improve performance reduce time, and handle large-scale data efficiently.

2.) what does scalability mean in real world applications?

Scalability means the system can handle increasing users, data, or workload without slowing down.

3.) Name companies that depend on scalable systems.

Amazon and Google.

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Grammar Correction:

1.) He didn't Knew the answer.
He didn't Know the answer.

2.) She is more smarter than him.
She is smarter than him.

3.) They has completed the task.
They have completed the task.

BLOCK-3: Communication Practice:

Explain a technical concept??

DBMS (Database Management System) is software used to store, manage, and retrieve data efficiently in an organized way. It acts like a bridge between the user and the database. The Real-world Example is student details with name, roll number, marks etc). They are stored in a database. Example MySQL, MongoDB, Oracle. Use cases are online shopping website (like Amazon) it stores product details, manages customer information, Tracks orders and payments etc...

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